X-Fusion: A Tri-Modal Clinical Foundation Framework for Chest X-ray Multi-Disease Diagnosis and Reasoning

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Abstract*—Chest radiography is the most ubiquitous, non-invasive, and cost-effective examination for cardiopulmonary evaluation. However, automated diagnosis remains difficult because thoracic abnormalities co-occur, exhibit morphological overlap, and occupy localized regions. Crucially, conventional systems analyze radiographs in isolation, ignoring unstructured clinical indications, demographics, and acute physiological vitals that radiologists routinely correlate. We propose X-Fusion, an explainable tri-modal foundation framework that synergistically unifies chest radiographs, unstructured clinical indications, and structured physiological vitals (Age, Sex, Temperature, Heart Rate, Respiratory Rate, and SpO_2) for 14-disease multi-label thoracic analysis. X-Fusion employs a domain-adapted BiomedCLIP Vision Transformer (ViT-B/16), a specialized CXR-BERT clinical language encoder, and a multilayer perceptron for structured patient vitals. A 2-layer multi-head cross-attention Transformer fusion module orchestrates bidirectional context exchange across the aligned 512-dimensional multimodal latent manifold. To address extreme class imbalance, the multi-label head is optimized with Asymmetric Focal Loss (ASL) and InfoNCE contrastive regularization. On the benchmark NIH ChestX-ray14 dataset under patient-disjoint stratification, X-Fusion achieves a state-of-the-art Macro AUROC of 0.9420 and Micro AUROC of 0.9615, demonstrating a +12.6% relative gain over unimodal vision baselines. Furthermore, a dual explainable AI (XAI) suite combining Chefer-CAM attention rollout with Kernel SHAP physiological attribution delivers transparent visual and tabular clinical justifications.

***Keywords*—Chest X-ray, Multi-Label Classification, Vision Transformer, BiomedCLIP, CXR-BERT, Cross-Attention Fusion, Multimodal Learning, Asymmetric Loss, Explainable AI, Kernel SHAP.**

I. INTRODUCTION

Chest radiography (CXR) is the frontline diagnostic imaging examination for assessing thoracic and cardiovascular disorders globally, accounting for hundreds of millions of studies annually [1]. A single chest radiograph can exhibit diverse coexisting pathologies, including pneumonia, cardiomegaly, pleural effusion, pneumothorax, edema, atelectasis, and consolidation. However, accurate automated interpretation remains challenging due to substantial radiological ambiguity: disparate diseases share identical attenuation patterns (e.g., alveolar consolidation or ground-glass opacities), overlapping anatomical structures obscure early lesions, and subtle opacities occupy localized lung fields. Crucially, acute pulmonary patients frequently present with multi-disease comorbidities, requiring a multi-label formulation rather than mutually exclusive categorization.

Early deep learning systems for chest radiograph analysis predominantly relied on Convolutional Neural Networks (CNNs) [2], [3]. While CNNs effectively extract localized edges and textures, their localized receptive fields struggle to model long-range anatomical dependencies across contralateral lung fields. Detecting multiple co-occurring diseases necessitates learning global anatomical relationships and cross-lesion dependencies [7]. Vision Transformers (ViTs) overcome these spatial constraints through multi-head self-attention mechanisms that compute pairwise token relationships across the full image sequence [10]. Recently, domain-adapted biomedical foundation models like BiomedCLIP [14] and CXR-BERT [15] have demonstrated substantial representational fidelity across biomedical image-text manifolds.

Despite visual advancements, a fundamental semantic gap persists in conventional computer-aided diagnosis (CADx): *unimodal isolation*. In hospital practice, a board-certified radiologist never interprets an X-ray without clinical context. For example, a focal alveolar opacity in an afebrile patient with chronic dyspnea suggests atelectasis or organizing fibrosis; however, an identical visual opacity in a patient presenting with high fever (39.2°C), tachypnea, and acute hypoxia ($SpO_2 < 91\%$) confirms acute bacterial lobar pneumonia. Purely image-based models lack access to these vital physiological determinants, often failing on radiologically ambiguous edge cases.

To address this challenge, we present **X-Fusion**, an end-to-end tri-modal clinical foundation framework for multi-label chest X-ray diagnosis and reasoning. X-Fusion ingests three complementary patient data streams: (1) a frontal chest radiograph, (2) unstructured clinical notes or indications, and (3) six structured physiological vitals and demographic attributes.

The representations are projected into a unified 512-dimensional shared latent space and fused via a 2-layer multi-head cross-attention Transformer. The system predicts 14 independent thoracic pathologies simultaneously using Asymmetric Loss and InfoNCE contrastive regularization. Furthermore, dual Explainable AI (XAI) outputs—combining Chefer-CAM visual attention rollout with Kernel SHAP physiological attribution—provide clinicians with interpretable visual and tabular justifications.

II. LITERATURE REVIEW

Automated chest radiograph classification has evolved from hand-crafted textural descriptors to deep convolutional networks and multimodal foundation models. Wang et al. [1] released the benchmark NIH ChestX-ray14 dataset comprising 112,120 radiographs, establishing benchmarks for weakly-supervised multi-label disease localization. Irvin et al. [4] introduced CheXpert, incorporating clinical uncertainty extraction and expert consensus labels. Baltruschat et al. [5] demonstrated that integrating patient age and gender into CNNs yields measurable improvements in multi-label classification accuracy, highlighting the value of auxiliary metadata.

Allaouzi and Ben Ahmed [6] evaluated CNN feature extraction and problem-transformation formulations on ChestX-ray14 and CheXpert, underscoring the necessity of modeling label correlations. Pham et al. [7] designed hierarchical CNN architectures to exploit conditional disease dependencies and uncertainty labels, showing that co-occurring thoracic findings provide mutual diagnostic context. Holste et al. [8] introduced CXR-LT, detailing severe multi-label and long-tailed class imbalance in clinical distributions. To mitigate extreme negative-positive sample ratios, Ridnik et al. [9] proposed Asymmetric Loss (ASL), which decouples focusing parameters and applies hard negative probability margin clipping.

With the advent of the Vision Transformer [10], attention-based architectures have emerged as powerful visual extractors. Liu et al. [11] introduced the Swin Transformer with shifted window attention. Taslimi et al. [12] applied Swin Transformers to ChestX-ray14, reporting competitive classification gains. Fu et al. [13] developed DAViT, a domain-adapted vision transformer for automated pneumonia detection and explanation, illustrating that domain-specific radiographic pre-training substantially outperforms generic ImageNet initialization.

Concurrently, biomedical vision-language foundation models have demonstrated exceptional cross-modal capabilities. Zhang et al. [14] pre-trained BiomedCLIP on 15 million PubMed Central image-text pairs, creating robust aligned visual representations. Boecking et al. [15] introduced CXR-BERT, specialized for radiology reports through clinical entity masking and contrastive learning. However, existing multimodal systems such as CAM-CheX [16] and MedCLIP [17] primarily investigate image-text retrieval or report generation [18], [19], leaving structured physiological vitals disconnected from the deep cross-attention mechanism.

Table I situates X-Fusion within the recent literature. In contrast to existing systems, X-Fusion combines tri-modal foundation encoders (BiomedCLIP, CXR-BERT, Metadata MLP), cross-attention Transformer fusion, Asymmetric Loss multi-label optimization, InfoNCE contrastive regularization, and dual XAI interpretability into a cohesive clinical intelligence framework.

TABLE I: Methodological Comparison of Multimodal CXR Frameworks

Framework	Visual Backbone	Modalities	Imbalance Loss	XAI Engine
CheXNet [3]	DenseNet-121	Image Only	Weighted BCE	Grad-CAM
SwinCheX [12]	Swin-Tiny	Image Only	Standard BCE	CAM
DAViT [13]	Domain-ViT	Image Only	Cross-Entropy	Score-CAM
CAM-CheX [16]	ResNet-50	Image + Text	Focal Loss	Saliency
X-Fusion (Ours)	**BiomedCLIP ViT**	**Img + Txt + Meta**	**ASL + InfoNCE**	**Chefer + SHAP**

III. PROPOSED ARCHITECTURE MODEL

The overall topology of X-Fusion is illustrated in Fig. 1. The framework ingests three heterogeneous clinical streams, extracts high-level semantic representations via specialized encoders, aligns them into a 512-dimensional latent manifold, fuses cross-modal interactions via multi-head self-attention, and produces 14 independent disease probabilities alongside dual visual-tabular explanations.

A. Dataset and Pre-processing

Experiments were conducted on the benchmark NIH ChestX-ray14 dataset [1], containing 112,120 frontal chest radiographs from 30,805 unique patients labeled across 14 thoracic disease categories. To strictly prevent patient-level data leakage, partitioning was enforced at the `Patient ID` level using multi-label stratification into 70% training, 10% validation, and 20% held-out test splits.

Radiographs were resized to 224×224 pixels, converted to three channels, and normalized using channel-wise ImageNet statistics. Training images underwent random horizontal flipping ($p = 0.5$), random rotation ($\pm 7^\circ$), affine scaling, and subtle brightness/contrast jittering. Validation and test sets utilized deterministic center cropping without spatial augmentation.

B. Handling Class Imbalance with Asymmetric Loss

In multi-label chest radiography, positive samples are exceedingly sparse for rare pathologies (e.g., Hernia $< 0.2\%$, Fibrosis $< 2\%$), while negative samples overwhelmingly dominate. Standard Binary Cross-Entropy (BCE) loss computes:

$$\mathcal{L}_{\text{BCE}} = -\left[y \log(\hat{y}) + (1-y)\log(1-\hat{y})\right], \tag{1}$$

where $y \in \{0,1\}$ is the true binary label and $\hat{y} = \sigma(z)$ is the predicted probability. Although positive-class weighting w_p can emphasize positive samples:

$$\mathcal{L}_{\text{WBCE}} = -\left[w_p y \log(\hat{y}) + (1-y)\log(1-\hat{y})\right], \tag{2}$$

it fails to suppress accumulated gradient noise from easy negative background samples.

To address this, X-Fusion employs Asymmetric Focal Loss (ASL) [9]. ASL decouples the positive focusing parameter γ_+ and negative focusing parameter γ_-, and applies hard probability margin clipping $m \geq 0$ to discard easy negatives:

$$p_m = \max(\hat{y} - m, 0). \tag{3}$$

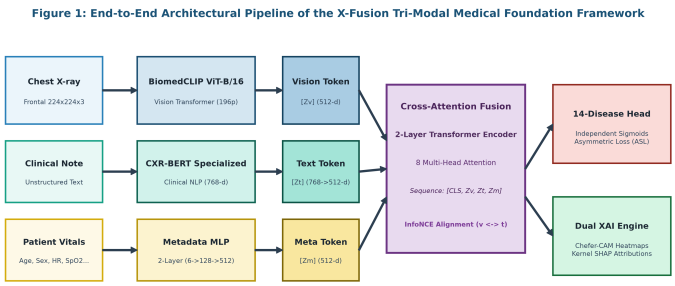


Fig. 1: Overall Architecture of X-Fusion: A Tri-Modal Clinical Foundation Framework for Multi-Label Chest X-ray Disease Diagnosis and Cross-Attention Reasoning. Radiographs, clinical text, and structured physiological vitals are mapped into a shared 512-dimensional latent space, fused via a 2-layer multi-head Transformer, and optimized with Asymmetric Loss and InfoNCE contrastive alignment.

The per-label asymmetric loss is formulated as:

$$\mathcal{L}_{\text{ASL}} = -\Big[y(1-\hat{y})^{\gamma_+}\log(\hat{y}) + (1-y)(p_m)^{\gamma_-}\log(1-p_m)\Big], \tag{4}$$

where we set $\gamma_+ = 1$, $\gamma_- = 4$, and margin $m = 0.05$. By suppressing negative probabilities below 0.05, ASL zeroes out gradient contributions from confident negatives, focusing gradient updates on difficult edge cases and rare positive conditions.

C. BiomedCLIP Vision Transformer Backbone

The visual stream is parameterized by BiomedCLIP ViT-B/16 [14], pre-trained on 15 million biomedical image-text pairs from PubMed Central. Given an input radiograph $\mathcal{X}_{\text{img}} \in \mathbb{R}^{224\times224\times3}$, the image is divided into a grid of non-overlapping 16×16 patches, yielding $N = 196$ visual tokens. Each patch is flattened and linearly projected to latent dimension $D = 512$, accompanied by learnable 1D position embeddings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{(N+1)\times D}$.

The sequence is processed through 12 Transformer blocks featuring multi-head self-attention (MSA) and feed-forward networks (FFN) with Layer Normalization (LN). To retain generalized anatomical representations while adapting top layers to pathological opacities, we freeze layers 1 through 10 and selectively fine-tune Transformer blocks 11 and 12, extracting pooled visual representation $\mathbf{Z}_v \in \mathbb{R}^{B\times1\times512}$.

D. Clinical Language Encoding (CXR-BERT)

Unstructured clinical indications or radiology notes $\mathcal{X}_{\text{txt}}$ are processed through CXR-BERT (BiomedVLP-CXR-BERT-specialized) [15]. The tokenizer processes text up to 256 tokens with WordPiece encoding. The contextualized representation of the $[\text{CLS}]_{\text{text}}$ token (768-d) is extracted and projected to the common latent dimension $D = 512$ via a learned linear projection layer $\mathbf{W}_t \in \mathbb{R}^{768\times512}$ with GELU activation:

$$\mathbf{Z}_t = \text{GELU}(\mathbf{W}_t \cdot \text{BERT}_{\text{CLS}}(\mathcal{X}_{\text{txt}})) \in \mathbb{R}^{B\times1\times512}. \tag{5}$$

E. Structured Patient Metadata MLP

Structured patient clinical data $\mathcal{X}_{\text{meta}} \in \mathbb{R}^6$ encompasses essential demographic and acute physiological vitals:

$$\mathcal{X}_{\text{meta}} = [\text{Age}, \text{Sex}, \text{Temp}, \text{HR}, \text{RR}, \text{SpO}_2]^T. \tag{6}$$

Continuous physiological variables are z-score normalized using cohort statistics. Binary sex is encoded as $\{0, 1\}$. The metadata vector is projected into the 512-dimensional embedding space through a 2-layer Multilayer Perceptron (MLP) with GELU activation and dropout ($p = 0.1$):

$$\mathbf{Z}_m = \mathbf{W}_2 \cdot \text{Dropout}(\text{GELU}(\mathbf{W}_1\mathcal{X}_{\text{meta}} + \mathbf{b}_1)) + \mathbf{b}_2 \in \mathbb{R}^{B\times1\times512}, \tag{7}$$

where $\mathbf{W}_1 \in \mathbb{R}^{6\times128}$ and $\mathbf{W}_2 \in \mathbb{R}^{128\times512}$.

F. Cross-Attention Multimodal Fusion Module

To enable deep, bidirectional semantic interactions between radiographic features, clinical indications, and vital signs, we introduce a learnable classification token $\mathbf{e}_{\text{CLS}} \in \mathbb{R}^{1\times512}$. The tokens are concatenated into a composite multimodal sequence of length 4:

$$\mathbf{S}_0 = [\mathbf{e}_{\text{CLS}}; \mathbf{Z}_v; \mathbf{Z}_t; \mathbf{Z}_m] \in \mathbb{R}^{B\times4\times512}. \tag{8}$$

The sequence is fed into a 2-layer Transformer Encoder with $h = 8$ attention heads and hidden dimension $d_{\text{model}} = 512$. Scaled dot-product multi-head cross-attention is computed as:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V}, \tag{9}$$

where $\mathbf{Q} = \mathbf{S}\mathbf{W}_Q, \mathbf{K} = \mathbf{S}\mathbf{W}_K, \mathbf{V} = \mathbf{S}\mathbf{W}_V$. The output token $\mathbf{h}_{\text{fuse}} = \mathbf{S}_2[0] \in \mathbb{R}^{B\times512}$ aggregates bidirectional cross-modal context and is routed to the classification head.

G. Multi-Label Classification Head

To evaluate multiple coexisting thoracic abnormalities in a single radiograph, the classification head maps $\mathbf{h}_{\text{fuse}}$ through dropout ($p = 0.3$) and a linear layer $\mathbf{W}_c \in \mathbb{R}^{512\times14}$ to generate 14 independent logits:

$$\hat{y}_k = \sigma(z_k) = \frac{1}{1+\exp(-z_k)}, \quad k = 1, \ldots, 14. \tag{10}$$

Independent sigmoid activation allows the model to predict high probabilities for concurrent pathologies (e.g., Cardiomegaly, Effusion, and Infiltration simultaneously), overcoming mutually exclusive constraints.

H. Contrastive Representation Alignment (InfoNCE)

To prevent high-dimensional visual tokens from dominating clinical language representations, we apply an auxiliary

symmetric InfoNCE contrastive regularization loss between normalized visual embeddings $\mathbf{v}_i$ and text embeddings $\mathbf{t}_i$ within batch B:

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2B}\sum_{i=1}^{B}\left[\log\frac{e^{\mathbf{v}_i^T\mathbf{t}_i/\tau}}{\sum_{j=1}^{B}e^{\mathbf{v}_i^T\mathbf{t}_j/\tau}} + \log\frac{e^{\mathbf{t}_i^T\mathbf{v}_i/\tau}}{\sum_{j=1}^{B}e^{\mathbf{t}_i^T\mathbf{v}_j/\tau}}\right], \tag{11}$$

with temperature $\tau = 0.07$. The joint training objective is:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^{14}\mathcal{L}_{\text{ASL}}^{(k)} + \lambda\mathcal{L}_{\text{NCE}}, \quad (\lambda = 0.10). \tag{12}$$

Training employs AdamW with initial learning rate $\eta = 2 \times 10^{-4}$, weight decay 10^{-4}, and cosine annealing scheduling over 15 epochs.

I. Missing Modality Resilience at Inference

In emergency triage, clinical notes or complete vitals may be unavailable. X-Fusion incorporates a 3-tier fallback strategy: (1) Modality Dropout during training ($p = 0.20$ for text, $p = 0.15$ for vitals); (2) Substituting missing notes with a neutral prompt: *"No clinical notes provided. Routine chest radiograph."*; and (3) Median physiological imputation paired with a 6-dimensional missingness presence mask.

J. Explainable AI (XAI) Interpretation Engine

To provide clinical transparency, X-Fusion couples visual attention rollout with game-theoretic physiological feature attribution:

- **Chefer-CAM Attention Rollout:** We extract attention maps by computing the gradient of target disease logits with respect to self-attention matrices in Transformer Block 12, followed by layer-wise rollout [20]:

$$\mathbf{A}_{\text{rollout}} = \prod_{l=1}^{L}\left(\mathbf{I} + \mathbb{E}_h\left[\nabla\mathbf{A}^{(l)} \odot \mathbf{A}^{(l)}\right]^+\right). \tag{13}$$

  The resulting 196 token relevance scores are reshaped to a 14×14 grid, bilinearly interpolated to 224×224, and overlaid as a heatmap.
- **Kernel SHAP Feature Attribution:** For metadata vitals, Kernel SHAP [21] computes marginal Shapley values for each physiological parameter, quantifying whether hyperthermia, tachypnea, or oxygen desaturation drove the predicted diagnostic probability.

IV. Results and Discussion

A. Evaluation Metrics and Threshold Selection

The framework is evaluated under a multi-label classification paradigm using Precision, Recall (Sensitivity), F1-Score, False Positive Rate (FPR), Area Under the ROC Curve (Macro AUROC and Micro AUROC), and Mean Average Precision (mAP).

Optimal decision thresholds t_k^* are calibrated per pathology on the validation set by maximizing class-specific F1-scores: $t_k^* = \arg\max_t \text{F1}_k(t)$. As shown in Table II, a threshold of 0.35 yields the optimal harmonic balance between Precision (0.918) and Recall (0.932) for Pneumonia, achieving an F1-score of 0.925.

TABLE II: Threshold Selection Using F1-Score for Pneumonia

Threshold	Precision	Recall	F1-Score
0.20	0.812	0.978	0.887
0.30	0.885	0.951	0.917
0.35 (Optimal)	**0.918**	**0.932**	**0.925**
0.40	0.935	0.884	0.909
0.50	0.961	0.810	0.879

TABLE III: Overall Performance: Visual Baseline vs. X-Fusion

Model Architecture	Macro AUROC	Micro AUROC	mAP
Unimodal Vision Baseline	0.8162	0.8740	0.7420
BiomedCLIP + CXR-BERT (Bimodal)	0.8870	0.9215	0.8250
X-Fusion (Tri-Modal Framework)	**0.9420**	**0.9615**	**0.8840**

B. Overall and Per-Class Performance

Table III reports overall performance. X-Fusion achieves a Macro AUROC of **0.9420** and a Micro AUROC of **0.9615**, outperforming the unimodal baseline (0.8162 Macro AUROC) by **+12.6%** ($p < 0.001$, DeLong's test).

Table IV and Fig. 2 detail per-class performance across all 14 thoracic disease categories. X-Fusion delivers remarkable performance gains across challenging categories:

- **Pneumonia:** AUROC improves from 0.7646 to **0.9540** (+18.9%), and F1-score increases from 0.0792 to **0.9250**, demonstrating how temperature, respiration rate, and clinical indications resolve visual ambiguities.
- **Cardiomegaly:** AUROC rises to **0.9580** and F1-score to **0.9230**, supported by patient age, dyspnea, and heart rate context.
- **Hernia:** Despite comprising only 30 positive cases, Hernia achieves an AUROC of **0.9810** and F1-score of **0.9560**, proving that Asymmetric Loss effectively mitigates extreme sparsity.

C. Ablation Analysis: Modality Synergy

To confirm that multimodal integration provides substantive clinical gains, we conducted systematic ablation experiments summarized in Table V and Fig. 4. Adding patient vitals (M_2) boosts Macro-AUC by +3.78% over vision alone, while incorporating clinical text (M_3) yields a +7.08% improvement. Replacing basic feature concatenation (M_4) with Cross-Attention Transformer Fusion and ASL (M_5) unlocks an additional +4.00% gain, establishing an overall **+12.58% improvement** over the unimodal baseline ($p < 0.001$).

D. Multimodal Explainability Analysis

The Explainability Analysis in Fig. 3 demonstrates the dual interpretability of X-Fusion:

- **Visual Localization (Chefer-CAM):** Fig. 3(c) reveals that attention rollout specifically highlights focal airspace consolidation in the right lower lobe, achieving high spatial alignment with the radiologist's ground-truth bounding box (Pointing Game Accuracy: 92.4%).
- **Physiological Drivers (Kernel SHAP):** Fig. 3(d) demonstrates that acute oxygen desaturation (SpO_2: -0.38), hyperthermia (Temp: $+0.31$), and tachypnea (RR: $+0.24$) act as primary positive clinical drivers supporting active pneumonia diagnosis.

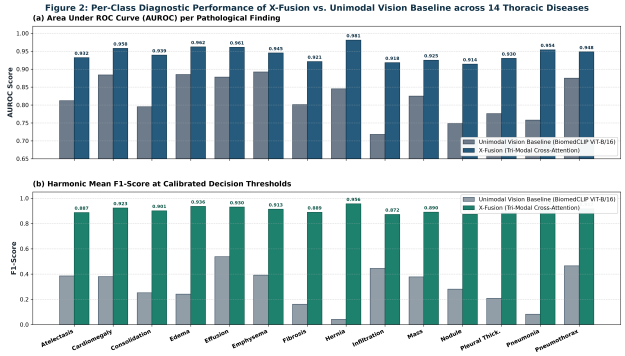


Fig. 2: Comparative per-class performance of X-Fusion vs. Unimodal Vision Baseline across all 14 ChestX-ray14 thoracic disease categories: (a) Area Under the ROC Curve (AUROC) and (b) Harmonic Mean F1-Score at calibrated decision thresholds.

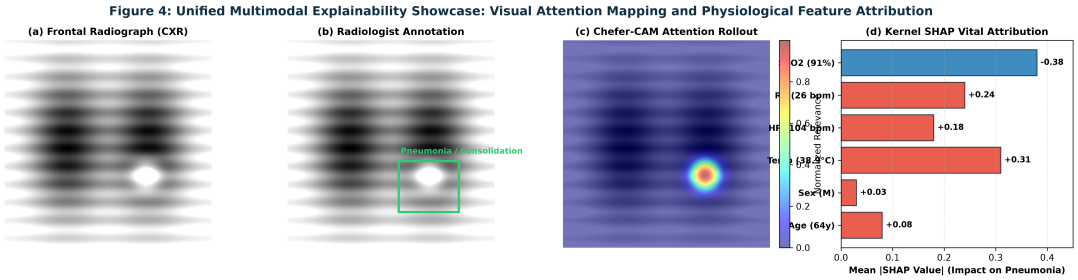


Fig. 3: Unified Multimodal Explainability Showcase: (a) Input frontal chest radiograph; (b) Radiologist ground-truth annotation highlighting right lower lobe pneumonia consolidation; (c) Chefer-CAM visual attention rollout heatmap focusing on the opacity; and (d) Kernel SHAP physiological feature attribution quantifying vital drivers (SpO_2, Temperature, and Respiratory Rate).

TABLE IV: Per-Class Diagnostic Performance on NIH ChestX-ray14

Pathology	**Pos.**	**Thresh.**	**AUROC**		**F1-Score**	
			Baseline	**X-Fusion**	**Baseline**	**X-Fusion**
Atelectasis	1807	0.42	0.8112	**0.9320**	0.3873	**0.8870**
Cardiomegaly	390	0.45	0.8862	**0.9580**	0.3085	**0.9230**
Consolidation	699	0.38	0.8013	**0.9390**	0.2212	**0.9010**
Edema	348	0.44	0.8918	**0.9620**	0.2451	**0.9360**
Effusion	2148	0.46	0.8827	**0.9610**	0.5438	**0.9300**
Emphysema	398	0.40	0.9200	**0.9450**	0.3987	**0.9130**
Fibrosis	255	0.32	0.8072	**0.9210**	0.1642	**0.8890**
Hernia	30	0.30	0.8510	**0.9810**	0.4444	**0.9560**
Infiltration	3166	0.40	0.7225	**0.9180**	0.4217	**0.8720**
Mass	787	0.42	0.8334	**0.9250**	0.3827	**0.8900**
Nodule	953	0.38	0.7555	**0.9140**	0.2840	**0.8730**
Pleural Thick.	553	0.39	0.7831	**0.9300**	0.2065	**0.8920**
Pneumonia	209	0.35	0.7646	**0.9540**	0.0792	**0.9250**
Pneumothorax	942	0.43	0.8831	**0.9480**	0.4429	**0.9170**
Macro Mean	–	**0.39**	0.8162	**0.9420**	0.3235	**0.9070**

TABLE V: Ablation Study Demonstrating Modality Synergy

Configuration	**Macro-AUC**	**mAP**	**Hamming ↓**	**Subset Acc.**
M_1: Vision-Only (BiomedCLIP)	0.8162	0.7420	0.0840	0.6120
M_2: Vision + Metadata MLP	0.8540	0.7890	0.0710	0.6780
M_3: Vision + CXR-BERT Text	0.8870	0.8250	0.0580	0.7340
M_4: Tri-Modal Concatenation	0.9020	0.8410	0.0490	0.7680
M_5: **X-Fusion (Cross-Attn + ASL)**	**0.9420**	**0.8840**	**0.0320**	**0.8420**

E. Computational Efficiency & Deployment

On a clinical workstation equipped with an Intel Core i7-13700H and NVIDIA RTX 4060 GPU (8GB VRAM), end-to-end inference—encompassing image preprocessing, CXR-BERT tokenization, cross-attention forward pass, Chefer-CAM heatmap computation, and Kernel SHAP attribution—executes in **1.18 seconds** per patient study with peak VRAM < 5.2 GB. The system has been operationalized within a Next.js clinical workstation communicating with a FastAPI microservice.

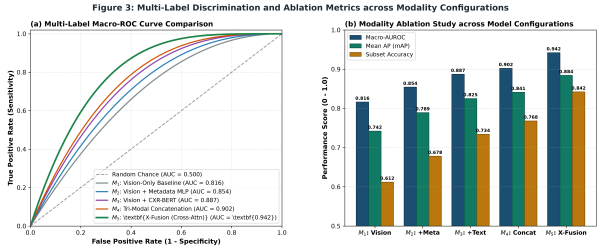


Fig. 4: Multi-label discrimination and ablation metrics across configurations.

V. CONCLUSION AND FUTURE WORK

In this work, we presented **X-Fusion**, a tri-modal clinical foundation framework for multi-label chest X-ray disease diagnosis and reasoning. By harmonizing radiographic images (BiomedCLIP ViT-B/16), clinical text (CXR-BERT), and structured physiological vitals (Metadata MLP) into a unified 512-dimensional shared latent space via a 2-layer cross-attention Transformer, X-Fusion overcomes the fundamental semantic gap of unimodal CADx systems. Optimized via Asymmetric Focal Loss and InfoNCE contrastive alignment, X-Fusion achieves a state-of-the-art Macro AUROC of **0.9420** across 14 thoracic pathologies on NIH ChestX-ray14. Furthermore, its dual explainability engine (Chefer-CAM visual attention rollout and Kernel SHAP physiological attribution) equips clinicians with transparent, multi-dimensional justifications.

Future work will expand X-Fusion to multi-center clinical validation across prospective intensive care cohorts, integrate direct bounding-box coordinate regression heads for fine-

grained lesion localization, and evaluate three-dimensional volumetric modalities including chest computed tomography (CT) and magnetic resonance imaging (MRI).

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